ZHOU, CONGHAO

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RESEARCH INTERESTS

dark matter · gravitational lensing · galaxy clusters · large scale structure
shear calibration · image simulation

EDUCATION

University of California, Santa Cruz

Expected 2027 Ph.D. in Physics
Thesis: *No Cluster Left Behind: Multiwavelength Cluster Analysis with All the Data*
Advisors: Tesla E. Jeltema, Alexie S. Leauthaud

Duke University

2020 B.A. in Physics with Highest Distinction
Thesis: *Intrinsic Alignment of redMaPPer Clusters in the Dark Energy Survey*
Advisor: Michael A. Troxel

SCIENCE PUBLICATIONS WITH MAJOR CONTRIBUTIONS

† STUDENT ADVISED OR CO-ADVISED

1. The Stellar-Mass Dependence of Feedback-Driven Outflows in DESI DR1 Star-Forming Galaxies
   arXiv:2611.02580 (2026), *submitted to Astrophys. J. Lett.*
   Y. Tang, Y. Luo, **C. Zhou**
2. Cluster finding with outskirt stellar masses and percolation
   arXiv:2608.19768 (2026), *submitted to Open J. Astrophys.*
   P. Avalos†, **C. Zhou**, T. Jeltema, K. Leidig, S. Xu et al.
3. Accurate Galaxy Cluster Shear and Mass Calibration for LSST with AnaCal
   arXiv:2606.21141 (2026), *submitted to Astrophys. J.*
   C. Zhou, X. Li et al.
4. Relationship between 2D and 3D galaxy stellar mass and correlations with halo mass
   J. Cosmol. Astropart. Phys. 10, 014 (2025) [arXiv:2502.05158]
   C. Zhou, A. Leauthaud, S. Xu, B. Diemer, S. Huang et al.
5. Forecasting the constraints on optical selection bias and projection effects of galaxy cluster lensing with multiwavelength data
   Phys. Rev. D 110, 103508 (2024) [arXiv:2312.11789], *Editors' Suggestion*
   C. Zhou, H.-Y. Wu, A.N. Salcedo, S. Grandis, T. Jeltema, A. Leauthaud et al.
6. The intrinsic alignment of red galaxies in DES Y1 redMaPPer galaxy clusters
   Mon. Not. R. Astron. Soc. 526, 323 (2023) [arXiv:2302.12325]
   C. Zhou, A. Tong, M.A. Troxel, J. Blazek, C. Lin et al.

TECHNICAL PUBLICATIONS WITH MAJOR CONTRIBUTIONS

1. Surface brightness profiles around massive galaxies from LSSTCam coadds
   LSST Technote SOTN-007 (2026)
   C. Zhou, T. Jeltema, A. von der Linden, L.S. Kelvin et al.

2. Surface brightness profiles around massive galaxies in LSSTComCam data
   LSST Technote SITCOMTN-165 (2025)
   C. Zhou, T. Jeltema, A. von der Linden, L.S. Kelvin et al.

CONTRIBUTING AUTHOR SCIENCE PUBLICATIONS

1. The Connection between Halo Assembly History and the Stellar Distribution II: A Radially Stratified Fossil Record of Past Halo Mass
   arXiv:2611.12470 (2026)
   S. Xu, S. Huang, **C. Zhou**, A. Leauthaud, B. Diemer

2. The Merian Survey: A Statistical Census of Bright Satellites of Sub-Milky Way-Mass Galaxies
   arXiv:2610.03981 (2026)
   L.S. Mandacarú Guerra et al.

3. Testing the Stellar Feedback-driven Breathing Mode in Low-mass Galaxies with Gas Kinematics
   Astrophys. J. 1009, 225 (2026) [arXiv:2510.17996]
   Y. Luo et al.

4. The Connection between Halo Assembly History and the Stellar Distribution I: The Central Stellar Mass–Richness Plane
   arXiv:2609.14488 (2026)
   S. Xu et al.

5. Probing Projection Effects in Optically-Selected Clusters with Velocity Dispersions Outliers
   arXiv:2609.08095 (2026)
   K. Graham et al.

6. The Vera C. Rubin Observatory Data Preview 1
   Astron. J. 171, 360 (2026) [arXiv:2603.23786]
   Vera C. Rubin Observatory Team et al.

7. DESI-DR1 3 × 2-pt analysis: consistent cosmology across weak lensing surveys
   Open J. Astrophys. 9 (2026) [arXiv:2512.15960]
   A. Porredon et al.

8. NSF-DOE Vera C. Rubin Observatory Observations of Interstellar Comet 3I/ATLAS (C/2025 N1)
   Astrophys. J. Lett. 1001, L35 (2026) [arXiv:2507.13409]
   C.O. Chandler et al.

9. Data Release 1 of the Dark Energy Spectroscopic Instrument
   Astron. J. 171, 285 (2026) [arXiv:2503.14745]
   DESI Collaboration et al.

10. Impact of projection-induced optical selection bias on the weak lensing mass calibration of galaxy clusters
    Phys. Rev. D 113, 063559 (2026) [arXiv:2510.00753]
    T.N. Nde et al.

11. Reaching for the Edge II: Stellar Halos out to Large Radii as a Tracer of Dark Matter Halo Mass
    arXiv:2511.10723 (2025)
    K. Leidig et al.

12. Cosmological Constraints from Dark Energy Survey Year 1 Cluster Lensing and Abundances with Simulation-based Forward-Modeling
    arXiv:2510.25706 (2025)
    A.N. Salcedo et al.

13. Dark energy survey year 3 results: Cosmological constraints from cluster abundances, weak lensing, and galaxy clustering
    Phys. Rev. D 112, 083535 (2025) [arXiv:2503.13632]
    T.M.C. Abbott et al.

14. The Outskirt Stellar Mass of Low-redshift Massive Galaxies Is an Excellent Halo Mass Proxy in Illustris/IllustrisTNG Simulations
Astrophys. J. 990, 127 (2025) [arXiv:2412.03406]
S. Xu et al.

15. Association between optically identified galaxy clusters and the underlying dark matter halos
Phys. Rev. D 112, 043517 (2025) [arXiv:2506.17526]
S. Cao et al.

16. Improving galaxy cluster selection with the outskirt stellar mass of galaxies
Phys. Rev. D 111, 123524 (2025) [arXiv:2410.20205]
M. Kwiecien et al.

17. Optical galaxy cluster mock catalogs with realistic projection effects: Validations with the SDSS clusters
Phys. Rev. D 111, 063502 (2025) [arXiv:2410.02497]
A. Lee et al.

18. Dark Energy Survey Year 3 results: Simulation-based cosmological inference with wavelet harmonics, scattering transforms, and moments of weak lensing mass maps. Validation on simulations
Phys. Rev. D 109, 063534 (2024) [arXiv:2310.17557]
M. Gatti et al.

19. Dark Energy Survey Year 3 Results: Deep Field optical + near-infrared images and catalogue
Mon. Not. R. Astron. Soc. 509, 3547 (2022) [arXiv:2012.12824]
W.G. Hartley et al.

CONTRIBUTING AUTHOR METHODOLOGY PUBLICATIONS

1. Getting Ready For Analysing Rubin Data: CLMM v2, The Updated LSST-DESC Cluster Weak Lensing Mass Modeling Library For Cosmology
arXiv:2610.37455 (2026)
M. Aguena, C. Combet, S. Fu, C. Payerne, M. Ricci et al.

2. OASIS: Observation-Aware Simulation-Based Inference via Distributional Matching
arXiv:2606.22572 (2026)
A. Farahi, **C. Zhou**, R. Vashistha

3. Building an Efficient Cluster Cosmology Software Package for Modeling Cluster Counts and Lensing
arXiv:2309.06593 (2023)
M. Aguena, O. Alves, J. Annis, D. Bacon, S. Bocquet et al.

COMPETITIVE FUNDING

2026–2027 **Chancellor's Dissertation Year Fellowship**, UC Santa Cruz, $33,000

2020 **Rodney I. McCormick Award**, Duke University, $1,500

TALKS

nov. 2026 **KIPAC, Stanford University and SLAC**, Tea Talk
No Cluster Left Behind: Multiwavelength Cluster Analysis with All the Data

oct. 2026 **Kavli IPMU, University of Tokyo**, Astro Seminar
No Cluster Left Behind: Multiwavelength Cluster Analysis with All the Data

aug. 2026 **Tsinghua University**, Department of Astronomy Seminar
No Cluster Left Behind: Multiwavelength Cluster Analysis with All the Data

jul. 2026 **Tsung-Dao Lee Institute**, Astronomy Seminar
No Cluster Left Behind: Multiwavelength Cluster Analysis with All the Data

jul. 2026 **Shanghai Jiao Tong University**, Department of Astronomy Seminar
No Cluster Left Behind: Multiwavelength Cluster Analysis with All the Data

may 2026	**South Pole Telescope Collaboration**, Cluster Working Group Meeting *No Cluster Left Behind: Joint Mass Calibration for DES and SPT Clusters*
dec. 2024	**University of Arizona**, Theoretical Astrophysics and Cosmology Seminar *Cluster Cosmology: Pixel to Object, CMB to Stellar Mass*
nov. 2024	**University of California, Davis**, Cosmology Seminar *Optical Cluster Cosmology: Selection to IA, Pixel to Object*
apr. 2024	**CMB-S4 Collaboration**, Cluster AWG Meeting *Calibrating Optical Selection Bias with Multiwavelength Data*
mar. 2024	**Brown University**, Center for the Fundamental Physics of the Universe Seminar *Cluster Cosmology Beyond Richness*
mar. 2024	**University of Pennsylvania**, Journal Club *Response Clustering Bias*
oct. 2023	**Tsinghua University**, Cosmology Seminar *Intrinsic Alignment in DES Y1 redMaPPer Clusters*
oct. 2023	**Stony Brook University & Brookhaven National Lab**, Cosmology Seminar *Constraining Optical Selection Bias in Cluster Lensing w/ Multiwavelength Data*

OBSERVING EXPERIENCE

10.5 effective nights	Dark Energy Spectroscopic Instrument, Mayall 4m at Kitt Peak
3 nights	DECaLS Intermediate Band Imaging Survey, Blanco 4m at CTIO

SCIENCE COLLABORATIONS

2026–present	Roman Science Collaboration
2024–present	Rubin Observatory Commissioning, In-Kind Contributor
2023–present	LSST Galaxies Science Collaboration
2023–present	LSST Informatics & Statistics Science Collaboration
2020–present	LSST Dark Energy Science Collaboration, Full Member since 2024
2020–present	Dark Energy Spectroscopic Instrument, Sponsored Member
2018–present	Dark Energy Survey, Student Member

PROFESSIONAL SOCIETIES

2026–present Sigma Xi, Member

2024–present American Astronomical Society, Student Member

2024–present American Physical Society, Student Member

OUTREACH

2023–2024 Starlight Over Street Light

2022 UC Santa Cruz Physics Outreach Taskforce

2022 Science Internship Program Mentor

2022 UC Santa Cruz Physics Mentoring Program Mentor

MEDIA COVERAGE

mar. 2025 **Astrobites**, Daily Paper Summary of Zhou et al. 2024 (Phys. Rev. D)
Less than meets the eye: measuring the bias in galaxy cluster observations

COLLABORATION SERVICE

jun. 2026–present LSST DESC Speaker's Bureau

2025, 2026 LSST DESC Collaboration Council Nomination Committee (Chair, 2026)

jul. 2022–present LSST DESC Mentoring Committee

apr.2022–may.2025 DES Early Career Scientists Committee (Chair, may 2024 – may 2025)

SEMINAR & MEETING ORGANIZATION

feb. 2027 SOC, LSST DESC 2027 Virtual Meeting

2025 Winter Physics Department Colloquium Committee, UC Santa Cruz

sep.2023–jun.2024 Organizer, CosmoGal Seminar, UC Santa Cruz

jul.2023–jun.2024 Organizer, BCG Seminar, Boise State University

jan. 2023 LOC, APS Conference for Undergraduate Women in Physics, UC Santa Cruz

aug. 2022 SOC, LSST DESC 2022 Chicago Meeting

REFEREEING

2024–present Astronomy & Astrophysics

TEACHING EXPERIENCE

2024 Spring	Application of Quantum Materials (Grader)
2024 Winter	Electricity, Magnetism, and Optics (TA)
2022 Fall	Quantum Mechanics (TA)
2022 Summer	Intermediate Physics Laboratory (TA)
2022 Spring	Advanced Astrophysics Laboratory (TA)
3 times	Introductory Physics Laboratory (TA)

November 12, 2026